

$$T(n) = 2T(n-1) + n2^{n-1} \rightarrow n > 0$$

$$\boxed{T(0) = 1} \rightarrow \text{Given}$$

$$u(n) = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases} \quad u(-1) = 0$$

$$u(n-1) = \begin{cases} 1 & n \geq 1 \\ 0 & n < 1 \end{cases}$$

At $n=0$

$$u(0) = 1$$

$$u(-1) = 0$$

$$\leftarrow T(n)u(n) = \underbrace{2T(n-1)}_{\downarrow} \underbrace{u(n-1)}_{\downarrow u(n-1)=0} + \underbrace{n2^n u(n)}_{\downarrow} + \underline{\underline{\delta(n)}}$$

$$\text{But } \boxed{T(0) = 1 = \delta(n)}$$

$$\boxed{T(n)u(n) = 2T(n-1)u(n-1) + n2^n u(n) + \delta(n)}$$